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Polyacrylic acid based plasma fractionation for the production of albumin and IgG: Compatibility with existing commercial downstream processes

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Abstract

Commercial fractionation of human plasma into immunoglobulin- and albumin-rich fractions is often initiated with sequential cold ethanol-based precipitation methods, which have changed little over the past 70 years. The required low temperature (−4 to −8°C) and high concentrations of ethanol 8–40% necessitate large-scale fixed processing lines, and major capital investment and operating costs. The resulting fractions are then further purified by ethanol based precipitation or chromatographic procedures to obtain the purified final product. Aqueous polyacrylic acid (PAA) based precipitation, which readily interfaces with existing downstream processing, could offer advantages with respect to cost, safety, environmental impact, and flexibility. Sequential precipitation with 7%, 12%, and 20% (w/v) solutions of PAA 8000 in the presence of a kosmotropic salt (sodium citrate) gave fibrinogen-, immunoglobulin-, and albumin-rich fractions with 80–90% yield and 64%, 55%, and 82% purity, respectively. Further purification of the IgG-rich precipitate by caprylic acid precipitation and anion exchange chromatography, achieved a target purity of >99%. This was also achieved for the downstream processing of the albumin-rich precipitate using a two-step ion exchange chromatographic procedure. This work shows that PAA precipitation can be used in place of cold ethanol precipitation to generate crude IgG and albumin fractions which can be purified to final products of acceptable purity.

KEYWORDS

albumin, fractionation, human plasma, immunoglobulin, polyacrylic acid

1 | INTRODUCTION

Human plasma is the source of many proteins for therapeutic use. Plasma fractionation is a multibillion dollar industry which exhibits significant ongoing growth (Marcucci, 2013). This growth is mainly due to increasing demand for intravenous and subcutaneous immunoglobulin

products, however, albumin demand remains strong (Goss & Curling, 2013; The Market Research Bureau, Inc., 2016). Plasma fractionation processes are still largely based on cold-ethanol fractionation (CEF) methods developed in the 1940s by Cohn et al. (1946) and subsequently improved by Kistler and Nitschmann (1962). The CEF process is based on the precipitation of proteins at various ethanol concentrations and

Abbreviations: CEF, cold ethanol fractionation; CM, carboxymethyl; DEAE, diethylaminoethyl; DoE, design of experiments; ELISA, enzyme-linked immunosorbent assay; ID, internal diameter; Ig, immunoglobulin; IEX, ion exchange chromatography; MW, molecular weight; OA, octanoic (caprylic) acid; PAA, polyacrylic acid; PAGE, polyacrylamide gel electrophoresis; pI, isoelectric pH; SDS, sodium dodecyl sulfate.

pH conditions at low conductivity and subzero temperatures. Today, while the manufacture of albumin remains predominantly a CEF process, the manufacture of IgG, in many cases involves the generation of a crude IgG precipitate followed by further chromatographic processing. Only CSL Behring in its Australian facility has adopted a completely chromatography based process for the manufacture of albumin and IgG (Bertolini, 2016; Bertolini, Goss, & Curling, 2013). This process starts with previously frozen plasma, which allows ease of collection and isolation of blood clotting factors.

The effectiveness of the CEF process cannot be questioned. The generation of crude precipitate intermediates has the advantage that it facilitates efficient shipping of material to other sites for further processing. Without CEF the shipping of plasma would require handling greater volumes and hence would be logistically more challenging and expensive. The major disadvantages of CEF are the use of ethanol and the need for subzero temperatures. While the use of ethanol has some cost and some flammability issues which must be considered, the requirement for complex refrigeration facilities, and the attendant electricity costs and the resultant carbon footprint, constitute the major disadvantage. Therefore alternative methods for the fractionation of human plasma that address the challenges of CEF are attractive for the industry.

The ability of polyelectrolytes to induce flocculation and precipitation has been researched and used industrially for many years (Cooper, Dubin, Kayitmazer, & Turksen, 2005; Singh et al., 2016). One polyelectrolyte precipitant of particular interest is polyacrylic acid (PAA). PAA (which has various common names such as Carbomer or Carbopol) is an abundant synthetic polymer often sold for dollars a kilogram. Due to its polycarboxylic and hygroscopic properties it has a history of use in pharmaceutical and cosmetic formulation, and drug delivery (Orwoll & Yong, 1999). As such analytical methods exist for its quantification (Sweett & Rolfe, 1958; Luke & Cleaver, 2014). PAA is considered biocompatible and safe (Final Assessment Report of the Safety of Carbomers, 1982) and it can be produced with narrow molecular mass distributions (Kawaguchi, Takahashi, Tajima, Hirose, & Ito, 1996). PAA tends to be almost fully protonated and charge neutral at $\text{pH} < 4$. Under such conditions the polymers may self-associate and precipitate out of solution.

Some specific studies on the use of PAA for precipitation of proteins follow. An early study showed that if serum was adjusted to $\text{pH} 4.6$, rendering albumin and many other proteins to have a net positive charge, protein precipitation could be elicited by gradual addition of polyacrylic acid (Wieland, Goldmann, Kern, Schultze & Matheka, 1953). Sternberg and Hershberger (1974) noted that PAA precipitation could be used with other protein mixtures tended to preserve enzyme activity. As in the case of other researchers they varied pH below target protein isoelectric pH (pI) to enhance electrostatic effects, and used PAA of relatively high MW (10^6 Da range) to take advantage of polymer bridging induced flocculation. Precipitates formed with lower MW PAA tended to be harder to form but easier to resolubilise. In the case of hemoglobin, optimal precipitation occurred at a PAA/protein ratio (w/w) of 0.1. Further addition of precipitant redissolved the precipitate. This phenomena is

common in polymer bridging induced aggregation but may require processes based on PAA induced precipitation to be optimized for each application with respect to pH , conductivity, polymer MW, and polymer concentration (Carlsson, Malmsten, & Linse, 2003; Cooper et al., 2005). McDonald, Victa, Carter-Franklin, and Fahrner (2009) studied the ability of polyvinylsulfonic acid and polyacrylic acid (PAA) to precipitate monoclonal antibodies. The pI of such proteins is often above 8, which suggests polyacids could be used to isolate antibodies from host cell proteins and other acidic contaminants. It was found that polyacid precipitation approached chromatography in selectivity (McDonald et al., 2009). However, it required optimization of precipitation conditions for each defined process to account for feed variables and individual monoclonal antibody properties. In addition the precipitates often required dilute solutions of altered pH to effect resolubilisation.

General Electric Healthcare (Van Alstine, Shanagar, Hjorth, & Lacki, 2010) filed a patent on polyacid based precipitation of various proteins including monoclonal antibody (1–20 g/L) with relatively low MW polyacid polymers (NaPAA8000) at 20°C involving adjustment of the solution to $\text{pH} 7.0$, and the addition of >50 mM of kosmotropic salts such as NaCitrate or NaPhosphate. Inclusion of a kosmotropic salt allowed the precipitates to be readily resuspended at high protein concentration in a variety of buffers suitable for direct follow on chromatography. Similar results were obtained at 4°C and 8°C . A follow on patent application (Van Alstine, Berg, Kjorning, & Shanagar, 2013) extended the above to the isolation of plasma precipitates enriched in fibrinogen and immunoglobulins.

The present study was undertaken to establish a PAA-based plasma protein precipitation process as a possible replacement for CEF. The work initially verified the sequential fractionation of plasma proteins with increasing concentrations of PAA and then explored the suitability of albumin and IgG enriched PAA precipitates to be further processed, in the same manner as CEF precipitates, via industrially used operations involving an octanoic acid based precipitation procedure and chromatography.

2 | MATERIALS AND METHODS

2.1 | Reagents

NaPAA (45% solution; MW 8,000 and 35% solution 15000), caprylic acid, and calcium phosphate were obtained from Sigma-Aldrich (St. Louis MO). General laboratory reagents including sodium chloride, trisodium citrate, sodium hydroxide, hydrochloric acid, sodium acetate, and ethanol were obtained from Merck (Darmstadt, Germany), Thermo Fisher Scientific (Rockford, IL), and Dominion Salt Ltd. (Mount Maunganui). Harborlite 900 was obtained from World Minerals (San Jose, CA) and 10 kDa MWCO dialysis tubing was obtained from Thermo Scientific. All plasma intermediate solutions were filtered through a $0.22\ \mu\text{m}$ membrane (Durapore, Millipore Corporation, Bedford, MA) before chromatography. Stock solutions used were 0.8 M NaCitrate $\text{pH} 3, 5, 7$, or 9 ; 0.8 M NaPhosphate $\text{pH} 7$, 5 M NaCl, NaPAA 8000 (45% w/w), and 15000 (35%, w/w).

2.2 | Chromatography resins

MacroPrep HQ resin was obtained from BioRad Laboratories (Hercules CA). Diethylaminoethyl (DEAE) Sepharose Fast Flow (FF) and carboxymethyl (CM) Sepharose FF resins were obtained from GE Healthcare Bio-Sciences AB (Uppsala, Sweden).

2.3 | Antibodies and enzyme linked immunosorbent assay reagents

Antibodies against the human plasma protein of anti-thrombin III, which were used for immunoelectrophoresis, were obtained from Sapphire Bioscience (Redfern, New South Wales, Australia). Enzyme linked immunosorbent assay (ELISA) kits for IgG, IgM, transferrin, α_2 -macroglobulin, ceruloplasmin, α_1 -antitrypsin, apolipoprotein A1, apolipoprotein B, haptoglobin, α_1 -glycoprotein, and fibrinogen were obtained from Bethyl Laboratories Inc. (Montgomery, TX), ICL (Portland, OR), AssayPro (St. Charles, MO), Abcam (Cambridge, UK) and GenWay Biotech. Inc. (San Diego CA). Antibodies and IgA standard for the IgA ELISA were obtained from DakoCytomation Denmark A/S (Glostrup, Denmark) and Calbiochem-Novabiochem Corporation (San Diego, CA). Antibodies against human serum albumin, IgG, IgA, IgM, and transferrin, which were used for nephelometric analysis, were obtained from Beckman-Coulter Inc. (Fullerton, CA).

2.4 | Polyacrylamide gel electrophoresis

A 10 μ l aliquot of the sample to be analyzed was added to 10 μ l of sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) loading buffer containing 50% (w/w) NuPAGE LDS Sample buffer (4 $\times$; Invitrogen, Carlsbad, CA) and 20% (w/w) β -mercaptoethanol and the mixture was reduced by heating at 70°C for 10 min. The samples were cooled and 10 μ l loaded onto a NuPage 4–12% Bis Tris gel (Invitrogen) and analyzed alongside 5 μ l of Precision Plus molecular weight markers (BioRad, Hemel Hempstead, UK). Electrophoresis was performed in NuPage MOPS SDS Running Buffer (20 $\times$) (Invitrogen) at 150 V for 1 hr. Protein bands were stained using GelCode Blue stain reagent (Thermo Fisher Scientific, Waltham, MA) with shaking over night before destaining in MilliQ water for 24 hr.

2.5 | Sequential fractionation of plasma proteins with polyacrylic acid

2.5.1 | Experiment 1

Initial precipitation experiments were performed with human plasma from volunteers (Blood Central Unit, Uppsala). It was not defatted or otherwise treated before experimentation. The precipitation studies

were performed at room temperature and involved mixing 1.5 ml of plasma with polymer solution and 50 mM NaCitrate, 50 mM NaCl, pH 7 to achieve a final PAA (NaPAA15000) concentration of 5%, 8%, and 10% (w/w) buffer solution and a total volume of 5 ml. The final total plasma protein concentration was 20 g/L. The mixtures were briefly vortexed and then allowed to stand vertically for 15 min. The tubes were centrifuged at 3,000g for 10 min (Eppendorf 5810R centrifuge). The recovered supernatant and precipitates were analyzed by SDS-PAGE.

2.5.2 | Experiment 2

More detailed experiments were undertaken to define optimum conditions for the precipitation of IgG. A full factorial design of experiment (DoE) based on 5 ml experiments and the procedure noted above was used (GE Healthcare Life Sciences, 2014). The effect of NaPAA15000 concentration (4–2%) and NaCitrate (60–220 mM) on the precipitation of plasma IgG was investigated. PAA and NaCitrate were dissolved in 200 mM NaCl.

Total protein analysis was by Biuret assay, and specific immunoglobulin analysis was by surface plasmon resonance based assay using a Biacore T1000. Results were verified using ImageQuantTL (GE Healthcare) analysis of reduced PAGE gel bands.

2.5.3 | Experiment 3

Conditions and results for the NaPAA8000 fractionation of plasma at 5°C were based on those outlined in Van Alstine et al. (2013). Frozen human plasma obtained from CSL Behring was thawed at 2–8°C and centrifuged at 3,910g for 20 min (4°C) to remove the cryoprecipitate fraction. The cryosupernatant fraction was divided into 12 portions and each adjusted to 50 mM NaCl and 50 mM trisodium citrate. Six of the samples were adjusted to achieve final PAA concentrations of 5%, 8%, 12%, 14%, 16%, and 20% (w/v). Each aliquot was adjusted to pH 7.0 with 1 M NaOH and 1 M HCl and then incubated at 2–8°C for 1 hr. Control samples of cryosupernatant were prepared as above but with the addition of the PAA solution substituted with distilled water. Following incubation each of the aliquots were centrifuged at 3,910g for 20 min. The supernatant samples were assessed for albumin, IgG and fibrinogen. Comparison of protein concentration in solution before and after precipitation was used to calculate % precipitation and $\ln K$, where K is the ratio of protein in precipitate to supernatant.

2.6 | Scale-up of polyacrylic acid fractionation of human plasma

Frozen human plasma was thawed at 2–8°C and centrifuged at 3,910g for 20 min (4°C) to recover the cryoprecipitate fraction, which was discarded. The cryosupernatant fraction was adjusted to 7% w/v

PAA, 50 mM NaCl, and 50 mM trisodium citrate. The sample was adjusted to pH 7.0 using 1 M NaOH and 1 M HCl and incubated at 2–8°C for 1 hr. The resultant fibrinogen precipitate was recovered by centrifugation at 3,910g for 20 min (4°C). Following removal of the supernatant the pellet was washed with a 7% w/v PAA solution containing 50 mM NaCl and 50 mM trisodium citrate. The washed precipitate was recovered by centrifugation at 3,910g for 10 min (4°C). The supernatant wash was removed and combined with the initial supernatant and the mass of the combined supernatant and wash sample recorded. The mass of the fibrinogen precipitate was recorded. The 7% PAA supernatant was adjusted to 12% w/v PAA, 50 mM NaCl, 50 mM trisodium citrate, and pH adjusted to 7.0 using 1 M NaOH and 1 M HCl before incubation for 1 hr at 2–8°C. The resultant IgG precipitate was recovered by centrifugation at 3,910g for 20 min (4°C). Following removal of the supernatant, the pellet was washed with 12% w/v PAA, 50 mM NaCl, 50 mM trisodium citrate solution, and processed in the manner described above.

The supernatant from the 12% PAA fractionation was adjusted as above to 20% (w/v) PAA, 50 mM NaCl, 50 mM trisodium citrate, and pH 7.0, then incubated for 1 hr at 2–8°C. The resultant albumin precipitate was recovered by centrifugation at 3,910g for 20 min at 4°C. Following removal of the supernatant the pellet was washed with a 20% w/v PAA solution containing 50 mM NaCl and 50 mM trisodium citrate, pH 7. The washed precipitate was recovered by centrifugation and processed as described above. A sample of each of the 7%, 12%, and 20% PAA precipitates was weighed and mixed with distilled water at a mass ratio of 1:5 and mixed vigorously to solubilize the proteins. The original plasma sample and solubilised precipitate samples were assessed for the levels of all the human plasma proteins noted above. Protein recovery and purity was calculated for each of the PAA precipitate samples.

2.7 | Downstream processing of IgG fraction

The IgG-rich 12% PAA precipitate (above) was processed using a method based on Stucki et al. (2008). Following precipitate resuspension in 220 mM sodium acetate solution (pH 4.8) the sample was treated with caprylic acid (14 g/kg of suspension) and incubated for 120 min at 18–25°C. The suspension was treated with calcium phosphate (0.3 g/g of caprylic acid), mixed for 60 min at RT and clarified with Harborlite 900 followed by centrifugation and filtration. The clarified material was adjusted to pH 4.0 and dialyzed against 5 mM sodium acetate using a 10 kDa MWCO membrane. The dialyzed sample was prepared for anion exchange chromatography by adjusting the pH to 6.5 using 0.2 M NaOH, and conductivity to 0.8 mS/cm by dilution in distilled water, and then filtered through a 0.22 µm membrane. The IgG intermediate sample was loaded on to a packed column containing MacroPrep HQ resin (1.6 cm ID × 17.5 cm) previously equilibrated with 10 mM sodium acetate, pH 6.2. The nonretained (flow through) fraction collected during the sample loading and postsample 10 mM sodium acetate, pH 6.2 wash was

collected. The purified IgG sample was analyzed for the 14 plasma proteins noted above.

2.8 | Downstream processing of albumin fraction using ion exchange chromatography

The albumin precipitate derived from precipitation using 20% PAA was further purified using ion exchange chromatography based on the method previously of Yap et al. (1993). Albumin has an approximate pI of 4.9. The 20% PAA precipitate was resuspended in 0.9% w/v NaCl and mixed for 60 min. The solution was dialyzed against approximately 10 volumes of 1 M NaCl using a 10 kDa MWCO membrane, followed by dialysis against approximately 10 volumes of distilled water, and then against approximately 10 volumes of 10 mM sodium acetate (pH 5.2). The dialyzed sample was adjusted to pH 5.2 and a conductivity of 0.6–1.0 mS/cm. The sample was clarified before chromatography by filtration through a 0.22 µm membrane. The albumin intermediate was loaded on to a DEAE Sepharose-FF packed column (1.6 cm ID × 17.5 cm) previously equilibrated with 10 mM sodium acetate (pH 5.2). The column was washed with 10 mM sodium acetate (pH 5.2) before the bound albumin was eluted using 25 mM sodium acetate (pH 4.5). The albumin eluate was adjusted to pH 4.5 and 2.0 mS/cm before being loaded on to a CM Sepharose-FF packed column (1.6 cm ID × 17.5 cm) previously equilibrated with 25 mM sodium acetate (pH 4.5). The column was washed with 25 mM sodium acetate (pH 4.5) before the bound albumin was eluted using 110 mM sodium acetate (pH 5.5). The purified albumin sample was analyzed for other plasma proteins as noted above.

2.9 | Downstream processing of albumin fraction using ethanol fractionation

The albumin precipitate derived from precipitation using 20% w/v PAA was also further purified using cold ethanol fractionation (CEF) based largely on the method of Kistler and Nitschmann (1962). The 20% PAA precipitate was resuspended in a solution containing 10 mM sodium acetate (pH 5.85) + 30% v/v ethanol and the pH adjusted to 5.85. The temperature of the mixture was reduced to –8°C and mixed for 60 min. The sample was clarified by centrifugation at 3,910g for 20 min (–8°C). The supernatant (containing albumin fraction) was collected and adjusted to pH 4.8. The temperature was reduced to –8°C and mixed for 60 min. The sample was clarified by centrifugation at 3,910g for 20 min (–8°C). The pellet (containing the albumin fraction) was resuspended in a solution containing 10 mM sodium acetate (pH 4.6) plus 10% v/v ethanol. The temperature was adjusted to –3°C and mixed for 60 min. The sample was clarified by centrifugation at 3,910g for 20 min (–3°C). The supernatant (containing the albumin fraction) was collected and retained for characterization of the proteins noted above.

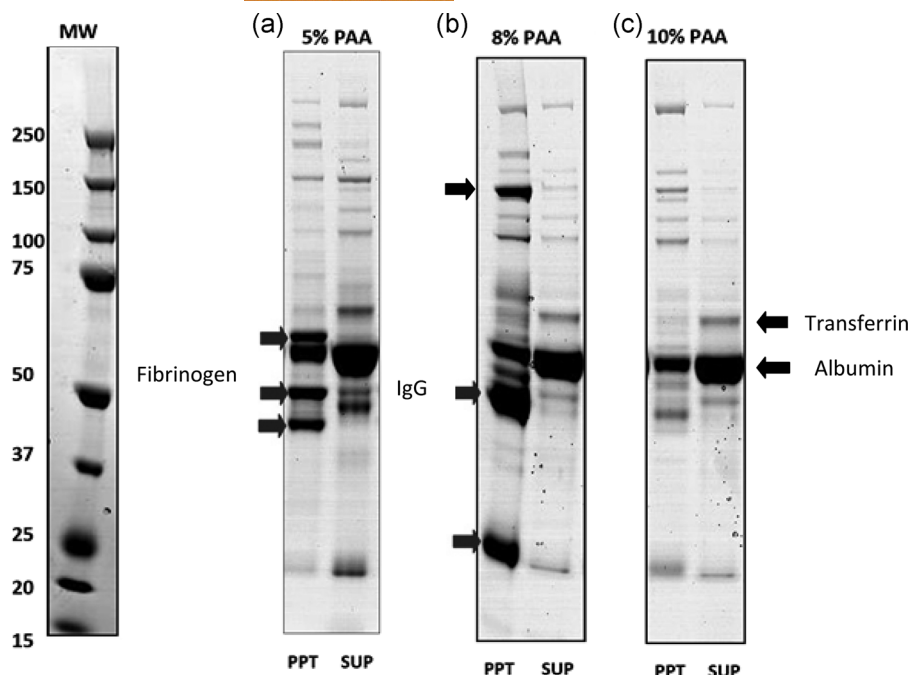


FIGURE 1 SDS-PAGE analysis of precipitate and supernatant fractions following sequential precipitation of human plasma proteins at PAA concentrations of 5% (w/w), 8% (w/w) and 10% (w/w). Precipitation was performed at room temperature, dilutions were performed using 50 mM NaCitrate, 50 mM NaPhosphate, pH 7.0. PAA, polyacrylic acid; SDS-PAGE, sodium dodecyl sulfate polyacrylamide gel electrophoresis

3 | RESULTS AND DISCUSSION

Figure 1 presents reduced PAGE analysis of precipitate and supernatant fractions generated from human plasma by NaPAA15000 precipitation. It can be seen that at 5% (w/w) PAA there is precipitation of fibrinogen followed by precipitation of IgG at 8% (w/w) PAA. Transferrin and albumin still remain in the supernatant at 10% (w/w) PAA.

Figure 2 presents the results for an experiment examining the relationship of IgG precipitation to PAA and NaCitrate concentration. It can be seen that the degree of IgG precipitation increases with increasing PAA concentration and this is enhanced with increasing NaCitrate. The NaCitrate solution used in this study contained 200 mM NaCl thus providing high ionic strength conditions. Thus the further effect of citrate addition on precipitation could be related to another mechanism than an increase in conductivity.

The use of less viscous solutions of NaPAA8000 were also examined for the fractionation of human plasma. PAA over range of 5–20% (w/v) yielded a similar precipitation pattern with respect to fibrinogen, IgG, and albumin, to those observed for NaPAA15000 and shown in Figure 1. The percentage of protein precipitated versus PAA concentration (Figure 3) revealed a classic sigmoidal shaped two-phase partition behavior (Albertsson, 1985). Differentiation of IgG and albumin precipitation was good, but at increased PAA levels the precipitation of fibrinogen and IgG converged.

Most polyelectrolyte-protein complexes have been studied at low conductivity with pH adjusted to promote electrostatic interaction between the polymer and protein target, with polymer bridging playing a key role when larger polymers (e.g., >100,000 MW) are used. The ability of polycarboxylic acids of lower MW such as 8,000 and 15000 to induce the precipitation of a variety of proteins from different feeds, such as plasma or clarified fermentation feed, at high

conductivity in the presence of >50 mM kosmotropic salts, suggests that under such conditions electrostatic interactions may play a lesser role in precipitate stabilization. Protein surface localization of PAA due to hydrophobic (as well as electrostatic) interaction (Carlsson et al., 2003; Fisher, 1987) may result in a charged neutralized polymer, which renders the protein complexes less soluble and prone to self-association. Under such conditions PAA precipitation can primarily be a function of PAA concentration, relatively independent of solution conductivity, and occur when the target proteins are net negatively charged. Given the excess of PAA in the present method, it may also be less dependent on target protein concentration. As such it may mimic precipitation by solvent or neutral polymers.

The findings shown in Figure 3 data were used to scale-up the PAA precipitation procedure to develop a three-step sequential plasma fractionation method. It was found that the fibrinogen-rich

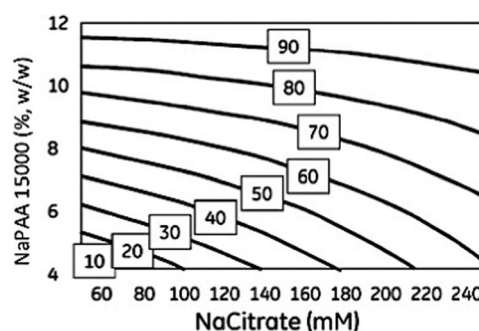


FIGURE 2 The effect of NaPAA and NaCitrate concentration on IgG precipitation (expressed as percentage IgG precipitated) from human plasma. NaPAA and NaCitrate solutions contained 200 mM NaCl and the final mixture was adjusted to pH 7.0 and incubated at room temperature

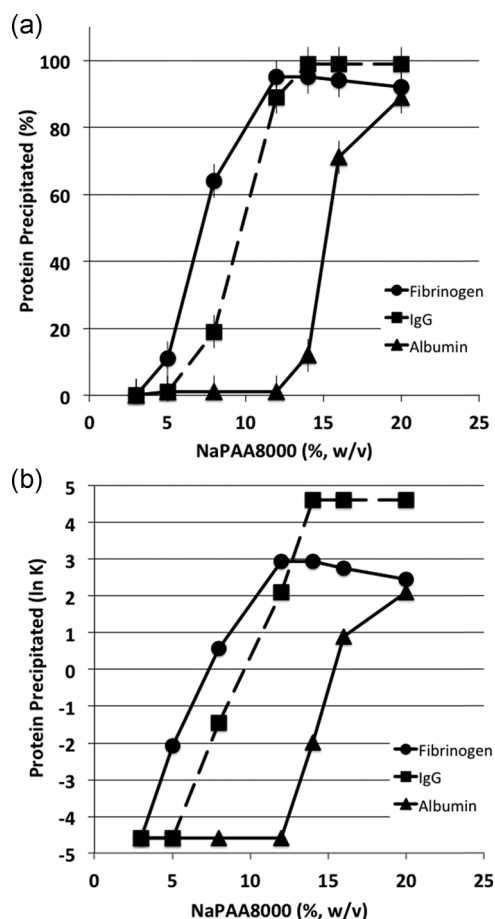


FIGURE 3 Precipitation of fibrinogen, IgG, and albumin from human plasma at 0%–20% with NaPAA8000. (a) % Recovery, (b) $\ln K$ (K = ratio of protein content in precipitate and supernatant). Buffer used was 50 mM NaCitrate, 50 mM NaPhosphate pH 7 and precipitation was performed at 5°C

fraction generated with 8% w/v, NaPAA contained significant amounts of IgG. To not compromise IgG recovery a 7% (w/v) polymer concentration was selected for fibrinogen precipitation. IgG precipitation was then performed by raising the NaPAA to 12% w/v. The majority of the albumin together with transferrin and some other contaminants was precipitated at 20% w/v NaPAA. The respective yields and percent-of-total-protein purities of target proteins in the precipitated fractions were fibrinogen (81 and 64%), immunoglobulin (89 and 55%), and albumin (81 and 82%; Table 1). The IgG fraction (55% target purity) contained appreciable albumin (11.8%), IgA (8.1%), α_2 -macroglobulin (9.1%), apolipoprotein B (4.7%), and fibrinogen (4.5%). As expected from CEF, the albumin fraction was contaminated with transferrin (7.0%), α_1 -antitrypsin (2.8%), and haptoglobin (2.2%; see Table 1). Given the fibrinogen fraction contained significant IgG (15%), and that a higher PAA concentration could increase the yield of albumin (Figure 3) it is likely that further process optimization would improve yield and purity.

The IgG and albumin enriched fractions were further processed by the procedures detailed in Figure 4 to obtain pure IgG and albumin. As previously reported by Van Alstine et al. (2013) the

precipitates derived from PAA fractionation could be stored frozen, and showed ease of solubilization when processed further using methods similar to those in current industrial use (Figure 4). They could often be dissolved in distilled water.

Treatment of the 12% PAA precipitate with octanoic acid (OA) increased IgG purity from 55 to 69% (Figure 5). OA treatment appeared to also enrich IgA and IgM. Following dialysis to adjust the conductivity of the sample and subsequent pH adjustment, the OA treated IgG was further purified to 99.9% by flow through ion exchange chromatography using MacroPrep HQ resin. The most abundant impurity after the OA precipitation and ion exchange (IEX) chromatography was IgA (0.05% of total protein; Figure 5).

Following resuspension of the 20% PAA precipitate and adjustment of the pH and conductivity, albumin was purified from 80.9% to 96.6% by IEX using DEAE Sepharose FF. IEX capture involving CM Sepharose FF then removed residual transferrin and apolipoprotein A1 resulting in 99.6% pure albumin (Figure 6).

The comparability between “5°C PAA8000 plus 50 mM kosmotropic salt” and “subzero CEF” is underscored by Figure 7 which indicates that under the conditions studied, every 1% of PAA, above the observed threshold of 3% which matches the x-axis intercept, has the same effect as 2.4% (v/v) ethanol ($R^2 = .998$).

Similarity between CEF and NaPAA based plasma protein fractionation is also supported by comparable levels of contaminants in the fractions. Several steps in CEF such as the 40% ethanol precipitation of Fractions IV and V are directed to enhancing albumin purity by precipitating such contaminants (e.g., haptoglobin, transferrin, and apolipoprotein A1). This suggested comparing if, instead of capture chromatography, the latter CEF process operations could also be used to further purify the 20% PAA albumin-rich fraction.

The direct similarity between ethanol precipitation and polyacrylic acid precipitation revealed in Figure 7 is in keeping with both methods being primarily dependent on reduced surface hydration of target proteins balanced against the proteins' inherent tendencies to remain in aqueous solution. It is therefore not surprising that the same order of proteins is seen precipitation of fibrinogen, IgG, and albumin by the uncharged hydrophilic polymer poly(ethylene glycol), although PEG-based precipitation may also require pH and conductivity (NaCl addition) manipulation to achieve effective yield and purity from the precipitation (Lee, Aishima, Nakai, & Sim, 1987). Similar protein precipitation trends are seen to varying degrees in other surface dehydration based separations such as hydrophobic interaction chromatography (Nfor et al., 2011) although the latter may be affected by secondary considerations such as protein geometry (Armstrong, Wenby, Meiselman, & Fisher, 2004). As such, the trend noted in Figure 7 is expected to hold for other plasma proteins. The slope appears to reflect differences in surface dehydration mechanisms such as charge-charge polymer-to-protein interactions playing a role during the initial stages of PAA-induced precipitation.

In a typical CEF process, addition of 40% (v/v) ethanol at pH 5.85 and -8°C , precipitates contaminant proteins allowing the albumin to remain in solution until pH is lowered to the pI of albumin. However

TABLE 1 Composition of fractions and protein recovery during polyacrylic acid fractionation of human plasma

Component	Composition (% of total proteins)			Recovery (% of original levels in plasma)		
	7% PAA precipitate	12% PAA precipitate	20% PAA precipitate	7% PAA precipitate	12% PAA precipitate	20% PAA precipitate
IgG	14.72	54.73	1.13	6.05	88.93	3.88
IgA	1.69	8.08	0.84	3.46	65.48	14.36
IgM	2.89	<0.01	0.03	12.48	0.03	0.99
Albumin	6.75	11.76	81.81	0.80	5.52	81.08
Transferrin	2.14	1.21	7.01	3.20	10.53	87.26
α_2 -Macroglobulin	2.18	9.14	0.68	4.81	79.93	12.46
Ceruloplasmin	0.56	0.93	0.57	8.14	53.22	68.72
Haptoglobin	0.30	1.85	2.18	1.04	25.29	63.04
Apolipoprotein A1	1.18	0.81	0.62	9.69	26.24	42.30
Apolipoprotein B	0.60	4.72	0.58	1.96	60.57	15.61
α_1 -Glycoprotein	1.32	1.09	1.10	7.72	25.19	53.60
α_1 -Antitrypsin	1.77	1.14	2.78	5.04	12.91	66.22
Antithrombin III	0.10	0.08	0.69	1.27	3.79	71.88
Fibrinogen	63.81	4.47	<0.01	80.71	22.32	<0.60

Note: The starting concentrations of fibrinogen, IgG and albumin were 1.72, 5.32, and 29.6 mg/ml, respectively. More details are provided in the Supporting Information tables.

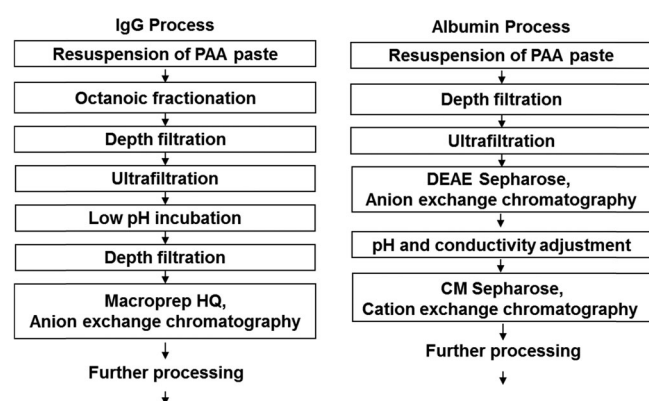
in trial studies the PAA precipitated and then resuspended albumin precipitated under typical Fraction IV conditions. The reason for this is presently unknown. It may be due to residual surface polymer or the protein becoming defatted in the PAA precipitation step. To achieve a condition where the Fraction IV proteins were precipitated while leaving the albumin in solution, the ethanol concentration was reduced to 30% v/v. Treatment of the 20% PAA precipitate with 30% v/v ethanol at pH 5.85 to precipitate the "Fraction IV" contaminant proteins increased albumin purity from 81.1 to 97.4%. This purity was similar to that post-DEAE IEX, and reflects significant depletion of transferrin, haptoglobin, apolipoprotein A1, α_1 -antitrypsin, IgG, α_2 -macroglobulin, IgA, apolipoprotein B, and ceruloplasmin. The content of both α_1 -glycoprotein and IgM increased moderately (for analytical data see Supporting Information Data). Following subsequent

precipitation of Fraction V at 30% (v/v) ethanol and pH 4.8, haptoglobin was reduced to 0.68% and α_1 -glycoprotein to 0.58% of total protein as albumin purity reached 98.6%. Although dual IEX capture chromatography is used in industry in larger (10 ton) processes, flow through IEX is also of interest.

The PAA 8000 used in this work may be cleared by the 10 or 30 kDa exclusion ultrafiltration membranes often employed in plasma fractionation. Further study is required to evaluate PAA clearance and residual levels, if any, in the final albumin and IgG fractions. A possible approach for this evaluation could involve fluorescent labeling of the PAA using 2-aminopyridine, which has previously been used for analysis of dextran sulfate and glycosaminoglycans (Araki et al., 2001; Kon, Takagaki, Kawasaki, Nakamura, & Endo, 1991).

The three step PAA fractionation increased plasma process volume 2.5-fold before the albumin precipitation. CEF processes typically show three-fold increases in volume. Further reduction in PAA process based volumes may be possible. Van Alstine et al. (2013) reported the precipitation works well even at a two-fold dilution, and it may be possible to obtain NaPAA stock solutions at >45%. In some cases it should be possible to combine the fibrinogen and IgG precipitation steps. This is often done in the plasma fractionation industry as established downstream operations are effective at removing fibrinogen from the immunoglobulin-rich fraction.

Polyacid-protein complexes have been studied for many decades and considerable information exists for the characterization of such complexes, the mechanism of their formation, their dissolution/resuspension, the design of unit operations compatible with larger scale batch and continuous complex formation, and comparative analysis of the economics of various processes (Fisher, 1988; Glatz,

**FIGURE 4** Processing steps for the purification of IgG and albumin from crude PAA precipitates. DEAE, diethylaminoethyl; PAA, polyacrylic acid

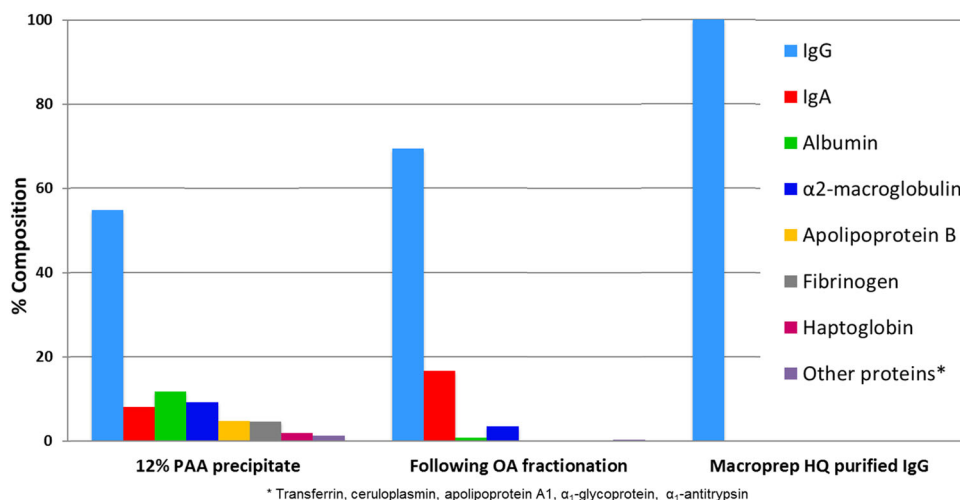


FIGURE 5 Composition of in-process intermediates generated by octanoic acid precipitation and anion exchange chromatography procedures during the processing of a crude IgG PAA precipitate. PAA, polyacrylic acid [Color figure can be viewed at wileyonlinelibrary.com]

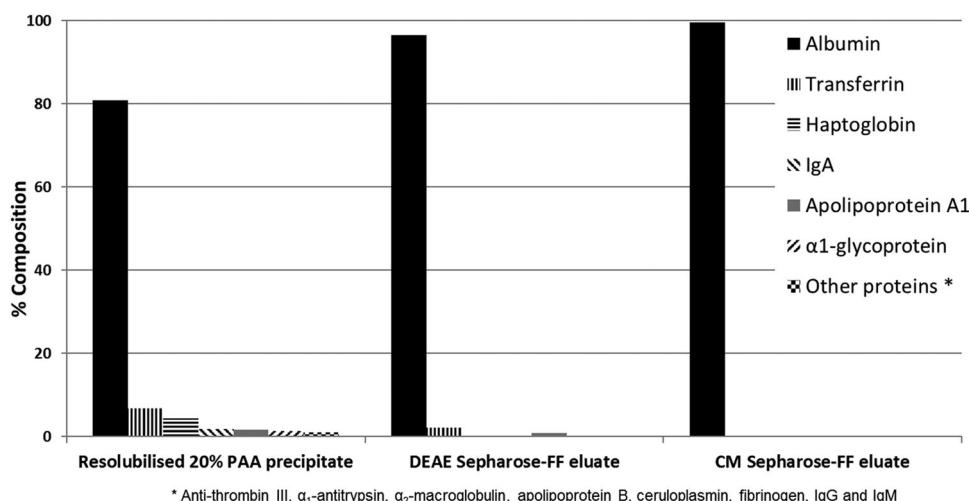


FIGURE 6 Composition of in-process intermediates generated during ion exchange chromatographic purification of a crude albumin PAA precipitate. PAA, polyacrylic acid

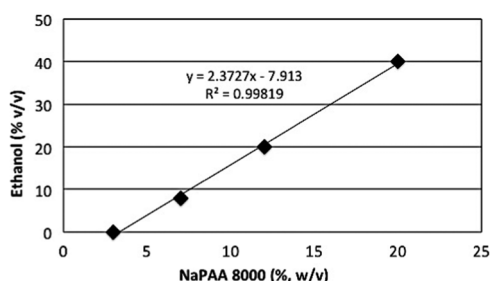


FIGURE 7 Comparison of concentrations of NaPAA8000 (w/v) which result in the precipitation of fibrinogen (7% NaPAA), IgG (12% NaPAA), and albumin (20% NaPAA) and ethanol concentrations 8, 20, 40 (% v/v), respectively, used in the Cohn process which result in the precipitation of these proteins. NaPAA8000 precipitation was performed at 5°C, in the presence of 50 mM NaCitrate, pH 7. Ethanol precipitation is performed at -6°C

Hoare, & Landa-Vertiz, 1986; Carlsson et al., 2003; Hammarschmidt, Hintersteiner, Lingg & Jungbauer, 2015). Though widely used industrially for applications such as water contaminant precipitation (Zhonghua, Campbell, & Somasundaran, 2001) and in formulations, PAA polymers do not appear to be used for industrial biomolecule separation. The main reasons for this may be the previous need, overcome in the present work, to optimize process variables such as pH and conductivity for each protein target and feed (McDonald et al., 2009). Polyacid based precipitation appears functionally similar to ethanol-based precipitation (Figure 7) yet without the challenges of solvent handling, subzero temperatures, or the requirement for target specific variation in unit operation pH.

Our results suggest that polyacid based precipitation involving relatively low MW polymers such as NaPAA 8000, in the presence of kosmotropic salts, is a selective and robust methodology for

precipitation of a wide variety of proteins from complex clarified feeds. As noted in this paper and related patent applications, essentially similar results are found at temperatures from 2 to 20°C, PAA MW 5000–15000, blood stabilized with different anticoagulants, and different kosmotropic salts (citrates or phosphates) and at varied pH (+/–1 unit). In agreement with Sternberg and Hershberger (1974) and Lombardi, Woitovich Valetti, Picó, and Valeria Boeris (2013) PAA plasma protein precipitates can be stored frozen and that the thawed precipitates can be readily resuspended. It has also been shown that the PAA precipitates, can be solubilised in different aqueous solutions, for example, distilled water (1 vol:1 vol) or phosphate buffered saline, and interface well with established downstream chromatographic processes previously optimized for use with cold ethanol precipitated protein fractions.

This work has also shown that a PAA based method facilitates the precipitation of fibrinogen, IgG, and albumin enriched fractions from human plasma, without the need for ethanol and subzero temperatures, and the resultant precipitates can readily be processed by existing plasma fractionation processes resulting in final products of comparable purity. Further work is warranted to formally establish process limits through the DoE format. Scale up of the process is then required to assess process robustness, process yield and provide sufficient final product for comprehensive final product analysis. The development of a convenient PAA assay will be required to allow confirmation of PAA removal through the course of the manufacturing processes.

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